Coastal Zone Extraction Algorithm Based on Multilayer Depth Features for Hyperspectral Images

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***Abstract*— The coastal zone is the most active natural area on the Earth's surface and has the most favorable resources and environmental conditions. Therefore, it is of great significance to conduct research based on the coastal zone. Hyperspectral remote sensing images have spatial and spectral dimensions that reflect the spatial distribution and can analyze the compositional information, which has been widely used for feature analysis and observation of ground objects. In this article, we propose a coastal zone extraction algorithm based on multilayer depth features for hyperspectral images (HSIs). The main contributions are as follows: 1) the Huanjing satellite hyperspectral coastal zone database is built for the first time, image composition is analyzed, and the noise removal algorithm is yielded; 2) 3-D attention networks that are capable of carrying spatial and interspectral information are proposed; and 3) A 3-D convolutional neural network (CNN) with squeeze and excitation network (SENet) tandem structure is proposed to fully exploit detailed information, and a multilayer feature extraction framework is built. We analyze four typical coastal zone patterns, and the experimental results show that our proposed algorithm can achieve coastal zone extraction with an average Kappa coefficient of 0.92, which is 0.06 higher than the mainstream algorithms. Our algorithm also shows good performance in complex environments. It provides a basis for further research on coastal zones.**

***Index Terms*— 3-D convolutional neural network (CNN), depth characteristic, hyperspectrum, multilevel, remote sensing, squeeze and excitation network (SENet).**

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I. Introduction

REMOTE monitoring refers to a technical approach to monitoring using remote sensing technology, with the major aim for ground cover, atmospheric, marine, and near-surface conditions. For remote monitoring technology, the satellite collection of environmental electromagnetic wave information on remote environmental targets is conducted to monitor the identification of environmental quality status. It has been reported as an advanced environmental information acquisition technology that is capable of acquiring large areas of simultaneous information in a short period of time and takes on critical significance in water quality monitoring methods. In marine remote sensing research, marine waters fall into Case 1 and Case 2 waters in accordance with their optical properties [1]. Case 1 waters refer to those where the optical properties of the water are primarily determined by chlorophyll and the detrital pigments that covary with it. In general, the optical properties of Case 2 waters are determined by suspended, colored soluble organic matter. Nearshore waters are directly affected by land-based sources carried by runoff from the coastal zone and estuaries and are classified as Case 2 waters [2]. Existing research based on nearshore waters can fall into research on constituents and research on regional extraction.

Existing research on the composition of water quality has placed stress on the following areas. Geological model-based methods are presented as follows: 3-D geological modeling of aquifer structure and groundwater [3], planktonic algae and water quality modeling [4], water quality variations in peri-urban and urban river watersheds modeling [5], multi-scale plankton water quality modeling [6], analysis of water quality variation, and environmental modeling based on a macro perspective [7]. Methods based on spectral response detection comprise microwave sensor response method [8], bacterial spectrum response method [9], water composition analysis method [10], ionization spectroscopy analysis [11], multidomain spectroscopy detection method [12], as well as microscopic imaging method [13]. Methods based on chemical properties cover chemical elemental analysis [14], water purification process analysis method [15], and bio composition method [16]. Mathematically based modeling: statistical method [17], principal component analysis (PCA) [18], neural network [19], convolutional neural network (CNN) [20], gradient boosting model (GBM) analysis and prediction of

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water quality [21], water quality monitoring indicator method [22], time variation analysis method [23], principal element content analysis method [24], as well as analysis of spatial and temporal variations [25].

Previous research on regional extraction has focused on the following areas. Image feature-based methods comprise texture features [26], color information [27], morphological characteristics [28], threshold characteristics [29], construction of fuzzy sets [30], energy characteristics [31], Bayesian modeling [32], multifractal analysis [33], graph algorithm [34], modified kernel K means [35], multiscale feature aggregation [36], normalized difference water index (NDWI) [37], and wave response method [38]. Depth feature-based algorithms: deep neural networks (DNNs) [39], DeepLab V3+ [40], separable residual segment network [41], as well as mask R-CNN [42], which have been employed to assess the effectiveness of different depth networks for water segmentation [43], DAU-Net [44], as well as the composite attention network (CAN) [45].

As a result of the above analysis, most of the existing research conducted thus far has conformed to IR and synthetic aperture radar (SAR) image spectra. The problems facing the research can be summarized as follows: 1) coastal zones contain different components, and a uniform model is difficult to build. 2) The boundary between waters and land is not significant, adding difficulty in extraction. 3) A single scale cannot accommodate different scales of coastal zone extraction.

A coastal zone extraction algorithm is proposed by fully exploiting a multispectral range of hyperspectral imagery: 1) The first hyperspectral coastal zone database of Huanjing stars is built, and a noise removal algorithm is proposed; 2) a 3-D attention network that is capable of carrying spatial and interspectral information is proposed; and 3) a 3-D CNN with squeeze and excitation network (SENet) tandem structure is proposed to fully exploit the detailed information.

The rest of this study is organized as follows. In Section II, the main framework of the algorithms in this study is presented. To be specific, Section II-A introduces the basic network structure. In Section II-B, the main noise patterns contained in hyperspectral remote sensing images are introduced, and a preprocessing model is built. Hyperspectral methods are adopted to acquire information, and hyperspectral data exhibit high dimensionality and can invert ground object characteristics. Thus, models should be built to analyze them in spatial and interspectral dimensions. Besides the winding coastline, the length and scale of the coastline is inconsistent and should be analyzed at multiple scales. In Section II-C, an attention model is built based on 3-D CNN network in spatial and interspectral dimensions to acquire information for feature extraction. To enhance feature mining at different scales, Section II-D builds the model from a multiscale perspective based on the SENet. Section III describes the dataset used for the experiments, and the experimental environment. Section IV summarizes the points of innovation and possible future work.

II. Proposed Algorithm

The near-shore water extraction algorithm is developed to conform to the practical application needs of near-shore waters for research, such that the characteristics of hyperspectral images (HSIs) with spatial and interspectral resolution are fully exploited. The exact process is illustrated in Fig. 1. The HSIs are first fed into an attention-based 3-D CNN network to mine the 3-D information regarding the images. Subsequently, SENet is developed to construct correlations between multilayered features, and nearshore water extraction is achieved by mining image information at different scales through multiscale-wide residual networks.

A. Introduction to Basic Networks

CNN network is highly flexible and capable of image feature extraction. There are several common types of CNN networks. 1-D CNN is a convolutional neural network based on spectral features, using the spectral features of each image element itself [46]. Assuming that each pixel is pure, each pixel in a hyperspectral remote sensing image can be represented by a set of spectral curves consisting of 1-D vectors. The 1-D vector corresponding to each pixel point in the HSI is directly used as the input matrix and the corresponding structure is designed to learn the spectral information of the HSI. However, the network does not make effective use of the information in the HSI data cube. 2-D CNN is a spatial feature-based convolutional neural network that use extracted correlations between image elements at different locations for HSI classification. It integrates spatial features into the network to enhance its performance [47]. The network compresses the spectral dimensional information, downscales the features in hundreds of spectral bands, and then convolves the downscaled data in 2-D to reduce the amount of operations. The above-described approach of compressing the spectral features before the extraction of the spatial features is capable of acquiring the spatial information of the pixels to be classified, whereas this acquisition process is at the expense of some of the spectral information. The spatial neighborhood of a pixel in HSIs can be processed with a 2-D CNN under the interdependence between neighboring pixels and the high probability that pixels in the spatial neighborhood fall into the identical class. The result is the 3-D CNN, which is a convolutional neural network based on spatial-spectral features [48]. It is difficult to eliminate the effects of pretzel noise by only placing a focus on spectral features. Besides, the detailed description of features in remotely sensed images cannot be described by focusing on spatial features alone. In contrast, 3-D CNN network considers multidimensional information and achieves better results.

In accordance with visual perception principles and extensive experiments, attention mechanisms can facilitate information utilization while reducing noise interference, and they have been extensively used in image segmentation. SENets is the more commonly used channel attention mechanism, divided into squeeze and excitation [49]. The feature map X is first input with a spatial and interspectral resolution of $C \times H \times W$ and compressed using the global average pooling (GAP) operation. The fully connected layer (FC) is then used to capture the correlations between the feature channels to obtain the incentive weights ω. Last, its output is multiplied channel-by-channel with the convolution output of

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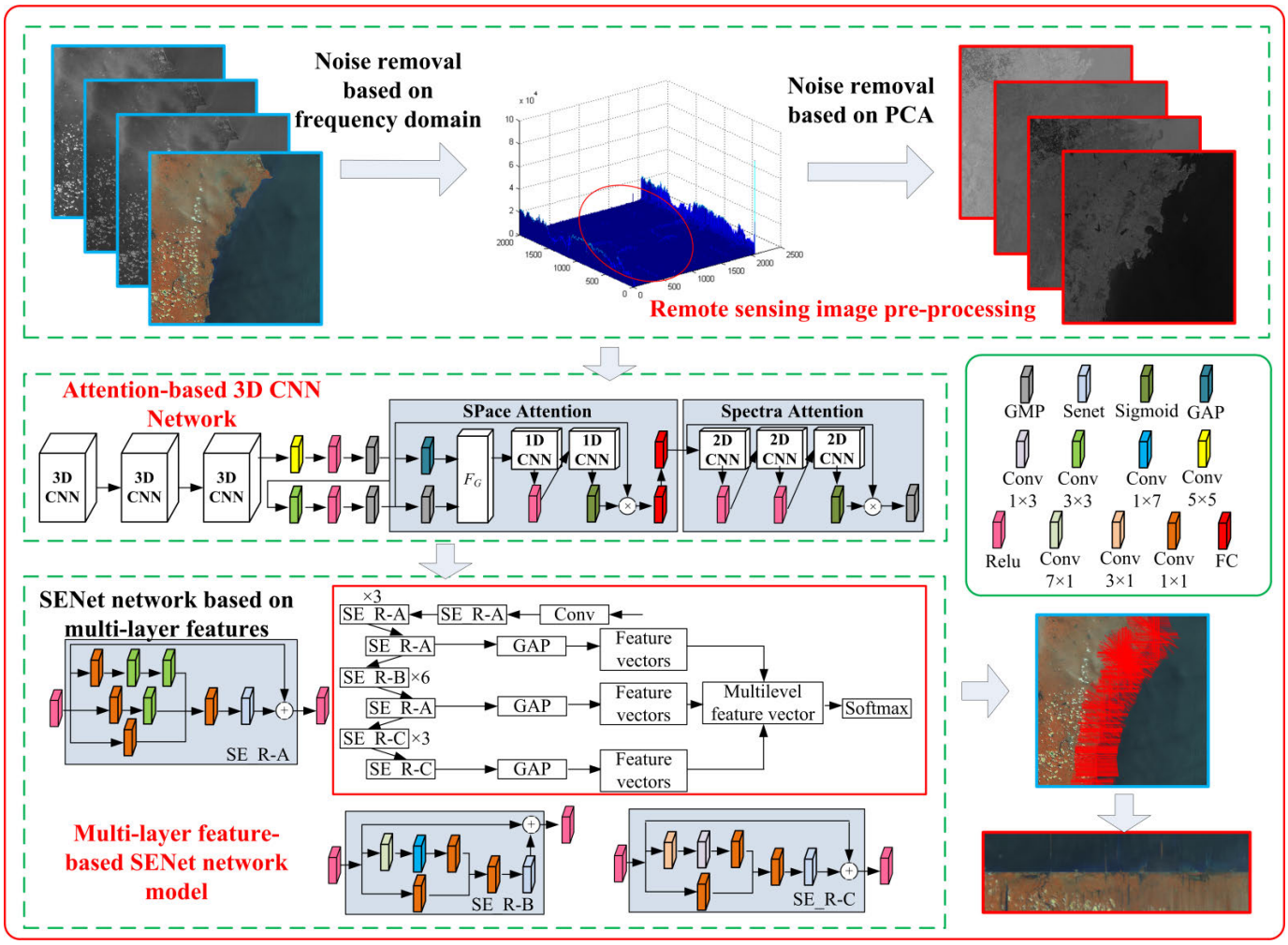


Fig. 1. Algorithm flowchart.

X to achieve channel rescaling of the feature map, as shown in Fig. 3. The output X_{out} of the SE module can be expressed as follows:

$$X_{\text{out}} = (\omega + 1) X_{\text{input}} \tag{1}$$

$$\omega = \text{sigmoid}\{\omega_2 * \text{ReLU}[\omega_1 * \text{Avg}(x)]\} \tag{2}$$

where ω_1 and ω_2 denote FC layer parameters.

To gain more insights into the image features, residual blocks are introduced to enhance performance [50]. Assuming that $H(x)$ denotes the optimal solution mapping of a deep neural network after input sample x, which is traditionally fit directly by an CNN

$$H(x) = x \tag{3}$$

while the deep residual network expects to fit the residual mapping

$$F(x) = H(x) - x. \tag{4}$$

Residual blocks that have been widely employed are presented in Fig. 3 and fall into the following categories: 1) base residual block. Two convolutional layers with a convolutional kernel size of 3 × 3 are used by spanning connectivity; 2) bottleneck residual block. A convolutional layer with a convolutional kernel size of 3 × 3 is added between two convolutional layers with a convolutional kernel size of 1 × 1, while a larger number of channels than the convolutional

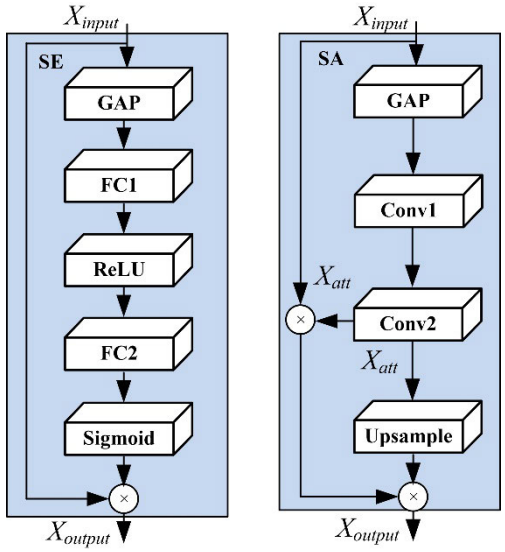


Fig. 2. Attention structure chart.

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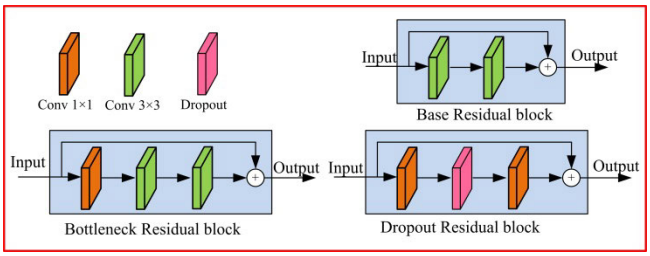


Fig. 3. Residual blocks.

kernel 3 × 3 is used to reduce the parameters of the model and increase the width of the network structure; and 3) dropout wide residual block. To enhance the test results and reflect the generalization of the wide residual network, a dropout mechanism is added in the middle of the two convolutional layers with a convolutional kernel size of 1 × 1. The improvement of the wide residual network on top of the residual block does improve the expression of the residual block.

B. Remote Sensing Image Preprocessing

Remote sensing images require a series of operations such as radiometric calibration and geometric correction before usable remote sensing image data can be obtained. Also, the acquisition process will inevitably introduce the effect of noise, resulting in inaccurate coastal zone extraction. To this end, we have analyzed the noise characteristics of remote sensing imagery and carried out research in the frequency domain and image composition. The fringe noises are generated by hardware and appear somewhat randomly [53]. Fringe noise is presented as a superposition of useful information and fringes in the spatial domain, and it is difficult to remove it in the spatial domain. Therefore, we construct a discrete Fourier transform (DFT) algorithm in the frequency domain. Moreover, in accordance with the idea of image dimensionality reduction, the PCA algorithm is adopted to extract the main components for image dimensionality reduction and enhancement.

The Fourier transform refers to a way of transferring an image from the spatial domain to the frequency domain for analysis. The 1-D DFT [51] is expressed as

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk}, \quad (k = 0, 1, \ldots, N-1) \tag{5}$$

$$W_N^{nk} = \exp\left(-j\frac{2\pi}{N}\right) \tag{6}$$

where $x(n)$ represents the input discrete digital signal sequence; W_N denotes the rotation factor; and $X(k)$ expresses the relative amplitude of the N discrete frequency points corresponding to the input sequence $x(n)$. The DFT and inverse discrete Fourier transform (IDFT) [52] corresponding to their extension to 2-D are expressed as follows:

$$F(\mu, \nu) = \sum_{x=0}^{M-1}\sum_{v=0}^{N-1} f(x, y) \exp\left[-j2\pi\left(\frac{\mu x}{M} + \frac{vy}{N}\right)\right] \tag{7}$$

$$f(x, y) = \frac{1}{MN}\sum_{x=0}^{M-1}\sum_{v=0}^{N-1} F(\mu, \nu) \exp\left[j2\pi\left(\frac{\mu x}{M} + \frac{vy}{N}\right)\right] \tag{8}$$

where μ, ν are frequency domain variables and x, y are spatial domain variables. By analyzing the twill noise in the frequency domain, as shown in Fig. 4, the response is shown to be prominent in the frequency domain. To do this, a threshold T is set, the part exceeding T is removed, and then IDFT is used to convert it into a spatial image to achieve the removal of the slant.

PCA is a method of projecting high-dimensional data into a low-dimensional space through a transformation matrix in order to reduce the dimension of the data [54]. The information is compressed to a finite sequence of frames by a PCA dimension reduction operation, where the amount of image information in the sequence tends to decrease, supported by literature and theory. The first few frames contain 98% of the information and the subsequent frames are considered noisy as the amount of information decreases, so we only take the first few frames to be calculated. The process is shown in Fig. 5 and the specific calculation steps are as follows.

Construct the sample matrix S, centering the p samples

$$S = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \cdots & \cdots & \cdots & \cdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}. \tag{9}$$

Calculate the sample covariance matrix

$$S = \frac{1}{n-1}\sum_{k=1}^{n}\left(x_{ki} - \bar{x}_i\right)\left(x_{kj} - \bar{x}_j\right). \tag{10}$$

The corresponding eigenvalues of the covariance matrix are λ_i and the eigenvectors. The eigenvalues with larger eigenvalues and the eigenvectors are selected to form a new eigenvector matrix. Then the corresponding principal component F_i is

$$F_i = Sa_i. \tag{11}$$

Calculate the cumulative contribution

$$\alpha_i = \lambda_i \Big/ \sum_{i=1}^{m} \lambda_i. \tag{12}$$

Calculate the sample principal components

$$F_i = a_{1i}X_1 + a_{2i}X_2 + \cdots a_{pi}X_p. \tag{13}$$

The above operations enable the image to retain its main components and denoise the image.

C. Attention-Based 3-D CNN Network

Due to the variability of the spectral response of the identical object in various bands, the spectral-spatial attention module is capable of automatically learning the importance of different bands and locations from the HSI, such that useful spectral-spatial information can be enhanced, and useless information can be suppressed.

Fig. 6 presents the spectral attention module. Given an intermediate feature map F, GAP, and global maximum pooling layer (GMP), are adopted to cascade the results to form a feature map F_G. The GAP is structurally regularized for the whole network to prevent overfitting and achieve spatial

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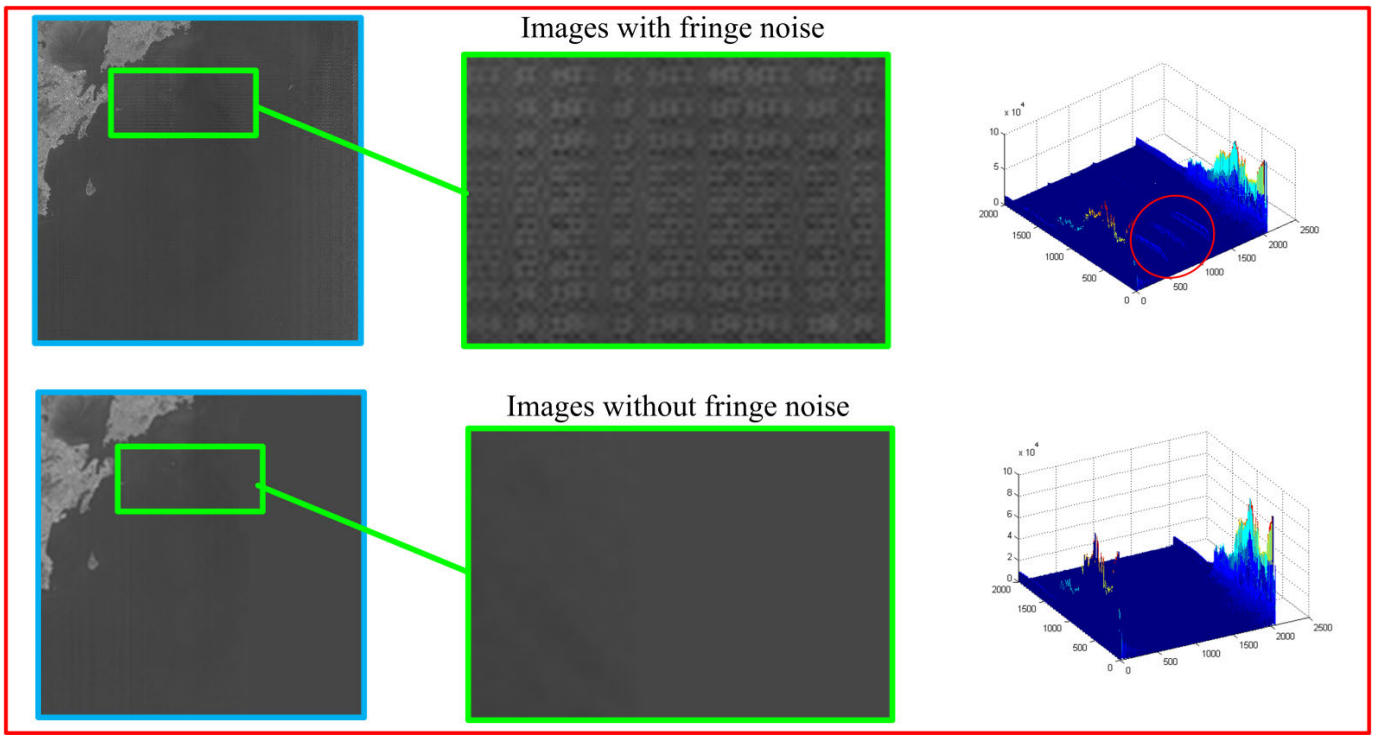


Fig. 4. Frequency image analysis with the fringe noise.

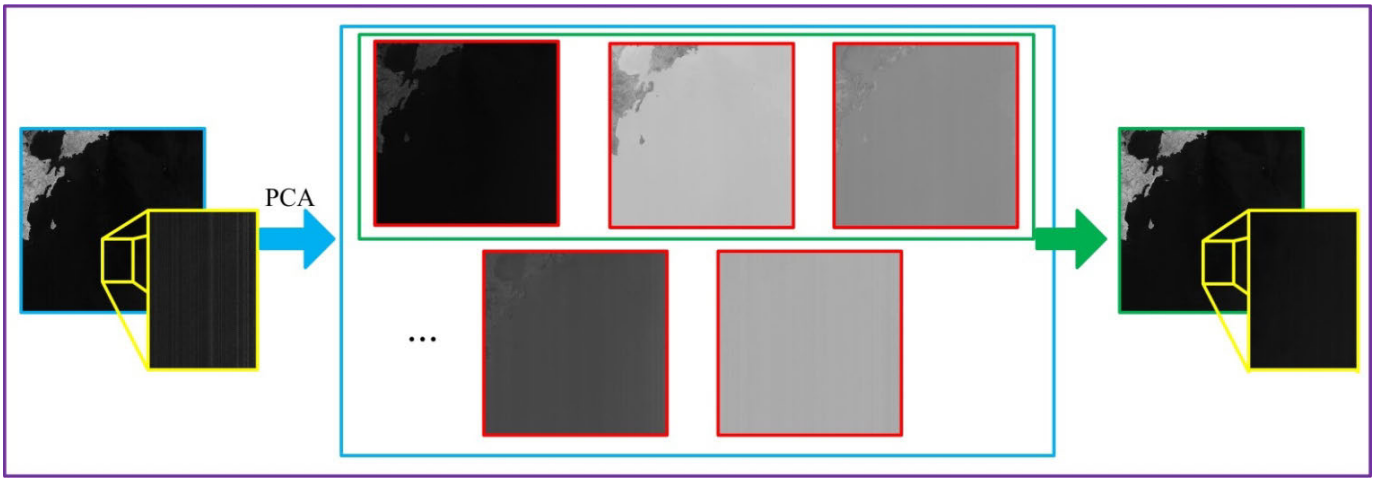


Fig. 5. PCA-based image denoising algorithm.

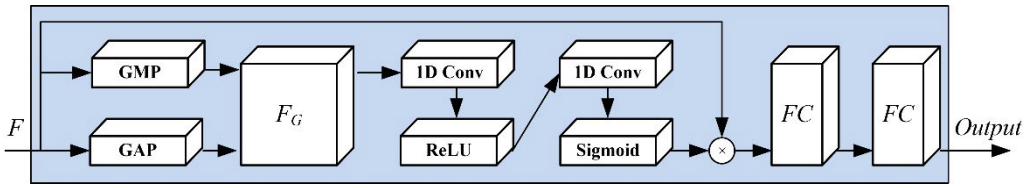


Fig. 6. Spectral attention model.

dimensional aggregation. Besides, the GMP complements the GAP with global features, which are sequentially passed through a filter and a 1-D CNN to generate a 64-channel spectral attention map F_{spe} of length m. Subsequently, F_{spe} is multiplied by F to generate the feature map F' expanded following the spatial dimension. Last, two FC layers are added to capture correlations between spectra, improving the differentiation of the refined spectral feature maps

$$F' = F \cdot F_{\text{spe}} \tag{14}$$

$$F_G = \text{Concat}(F_{\text{GAP}}, F_{\text{GMP}}) \tag{15}$$

$$F_{\text{spe}} = \text{sigmoid}(\text{Conv}(\text{ReLU}(\text{Conv}(F_G)))). \tag{16}$$

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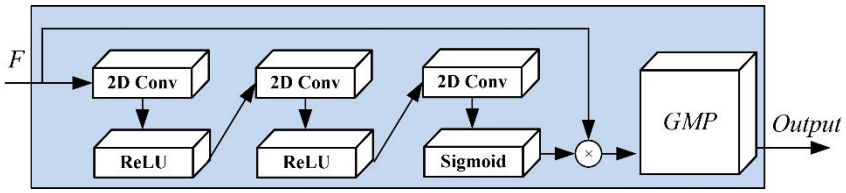


Fig. 7. Spatial attention model.

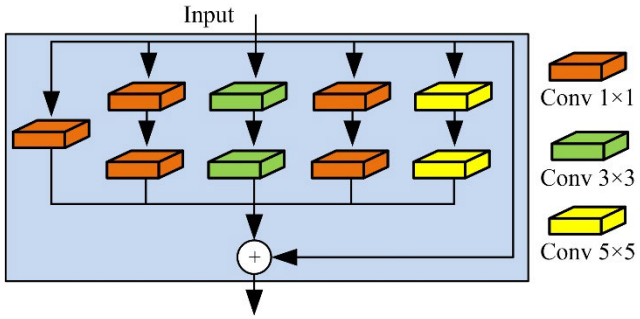


Fig. 8. Multiscale wide residual learning block.

The spatial attention module is presented in Fig. 7. The feature map F_C is obtained with 1×1 convolution, and the 2-D CNN with filters 1×1, 5×5, and 3×3 is constructed to obtain the spatial attention map F_{Spa}. Subsequently, F_{Spa} is expanded to F' through channel dimension. Last, the output is made fixed size with the maximum pooling layer

$$F' = F \cdot F_{\text{spa}} \tag{17}$$

$$F_C = \text{ReLU}(\text{Conv}(F)) \tag{18}$$

$$F_{\text{spa}} = \text{sigmoid}(\text{Conv}(\text{ReLU}(\text{Conv}(F_C)))). \tag{19}$$

Tandem structures are developed to fuse spectral and spatial attention modules, calculate weights, and enhance information.

D. Multilayer Feature-Based SENet Network Model

We refer to the Inception V2-Resnet network structure, as shown in Fig. 8. The SENet modules are embedded into Inception-Resnet-A, Inception-Resnet-B, and Inception-Resnet-C, respectively, while the multilayer features of the network are fused.

The specific parameters of the network structure are as follows:

1) The number of convolutional kernels in the first convolutional layer is 16, the size of the convolutional kernels is 3×3, and the step stride is 1.
2) In the SE-Inception-Resnet-A module, the first three convolutional groups are entered, in which the number of convolutional kernels is 16 and the step stride is 1. Next, the three convolutional groupings are merged and fed into a 1×1 convolutional layer with 32 convolutional kernels and a step stride of 1. This module is available in a total of three.
3) In the SE-Inception-Resnet-B module, two convolutional groupings are first entered, in both of which the number of convolutional kernels is 16 and the step stride is 1. Second, the two convolutional groupings are combined and fed into a 1×1 convolutional layer with 64 convolutional kernels and a stride size of 1. The total number of this module is six.
4) In the SE-Inception-Resnet-C module, two convolutional groupings are first entered, in both of which the number of convolutional kernels is 16 and the step stride is 1. Second, the two convolutional groupings are combined and then fed into a 1×1 convolutional layer with 128 convolutional kernels and a stride size of 1. This module is available in a total of three.
5) The feature maps output after three SE-Inception-Resnet-A, six SE-Inception-Resnet-B, and three SE-Inception-Resnet-C are pooled globally on average, respectively.
6) Since the higher and intermediate layers will still have some information, the feature vectors obtained by global averaging pooling of the feature maps of the lower, intermediate, and higher layers are feature fused to form more expressive feature information.
7) The feature vectors are fed into *Softmax* for classification.

The output of the deep supervised (DS) strategy is used in the network training process for one loss calculation and the loss of each lateral output is reflected in the final loss function, which is used to supervise the training process of the network. The calculation formula is defined as follows:

$$F = \sum_{m=1}^{M} W_s^{(m)} l_s^{(m)} + W_a l_a \tag{20}$$

$$l = -\sum_{(i,j)}^{(H,W)} \left[P_{G(i,j)} \log P_{S(i,j)} + \left(1 - P_{G(i,j)}\right) \log\left(1 - P_{S(i,j)}\right) \right] \tag{21}$$

where $l_s^{(m)}$ denotes the loss of inside output; l_a expresses the total final output loss; W represents the weight. $P_{G(i,j)}$ and $P_{S(i,j)}$ are the true and predicted values at coordinates (i, j), respectively. The network convergence is achieved by continuously learning the network weight parameters to minimize the value of l during training.

The capacity and feature discrimination of the model architecture is increased with the number of layers of the network, whereas the gradient dispersion problem occurs after a certain level of depth is reached. To address this problem, based on the inception module idea, we appropriately add branches with wide residual blocks to achieve multiscale feature learning. Furthermore, to reduce the parameters of the network model, instead of using larger convolutional kernels for the newly

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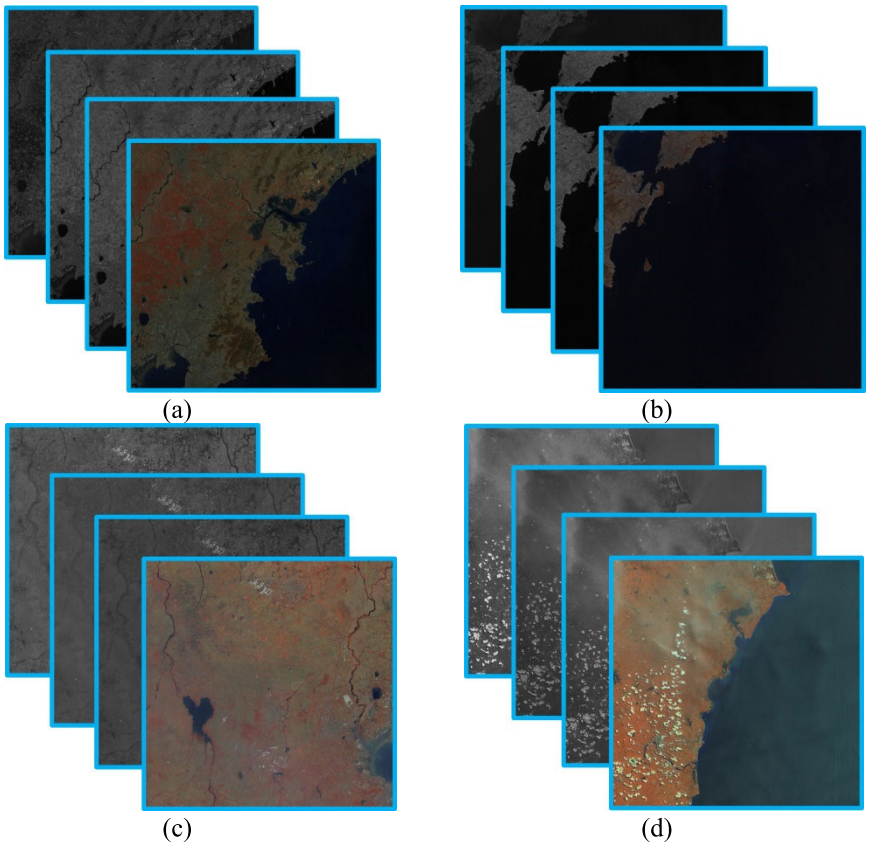


Fig. 9. Data presentation. (a) High Earth orbital satellite observation images. (b) Low Earth orbital satellite observation images. (c) Images of inland lakes in Qingdao. (d) Images of the coastal zone around Hainan.

added parallel connected convolutional layers, 1×1, 3×3, and 5×5 convolutional kernels are used to form a multiscale wide residual network, with corresponding Batch Norm and ReLU layers preceding each convolutional layer. The wide residual module has six branches (see Fig. 8). To improve the expressiveness of the network, it contains 1×1 convolution except for the branches connected by shortcut. The reason for the above result is that the 1×1 convolution changes the number of channels while adding feature transforms and nonlinear transforms. The total number of layers in the wide residual module reaches 16.

III. Experiment Analysis

Huanjing HSIs with an image resolution of 1985×2047 and a spectral band count of 100 are employed in this study, and the data are presented in Fig. 9. For a total of 98 HSIs, the coastlines in the images are annotated as a gold standard. The ratio of training to testing reaches 1:1. Besides, the amount of data will continue to increase. The block is 1000 randomly selected hyperspectral data at a learning rate of 0.01. The model is trained by calculating the loss function.

Fig. 9(a) and (b) present the HSIs taken at different moments at the identical location, which correspond to different spatial resolutions. Fig. 9(c) is an image of the inland lakes of Qingdao. Fig. 9(d) is an image of the coastal zone around Hainan. The fact that the data acquired are 16-bit makes the images appear unevenly bright and dark during the mapping to 8-bit.

Spectral curves are obtained by spectrometers. Spectral data can be considered as 3-D data, and each point in its space can be viewed as a spectral curve. This curve is the basis for analyzing different ground objects. Spectral curves of identical ground objects have a certain similarity. Based on the large amount of available data, we have extracted the spectral curves of the more typical objects shown in Fig. 10. The response curve for land is shown in Fig. 10(a), with a flatter response up to band number 50 and subsequent enhancement. Fig. 10(b) presents the water response curve, with the response being stronger up to band number 50 and smoother thereafter. The coastal zone response curve is shown in Fig. 10(c), with the integrated land and water curve features, showing a convex image overall. The response curve for the cloud is shown in Fig. 10(d), with high overall response values. From the above analysis, an experimental database is constructed and the coastal zone areas are marked by manual means as a gold standard to verify the effectiveness of the algorithm.

A. Algorithm Preprocessing Effects

As depicted in Fig. 11, the fast Fourier transform (FFT) image containing the slanting noise is presented, showing the raised state (red circle). In contrast, normal images show flat areas, for which a threshold T is set to invert the frequency domain image to achieve skew removal.

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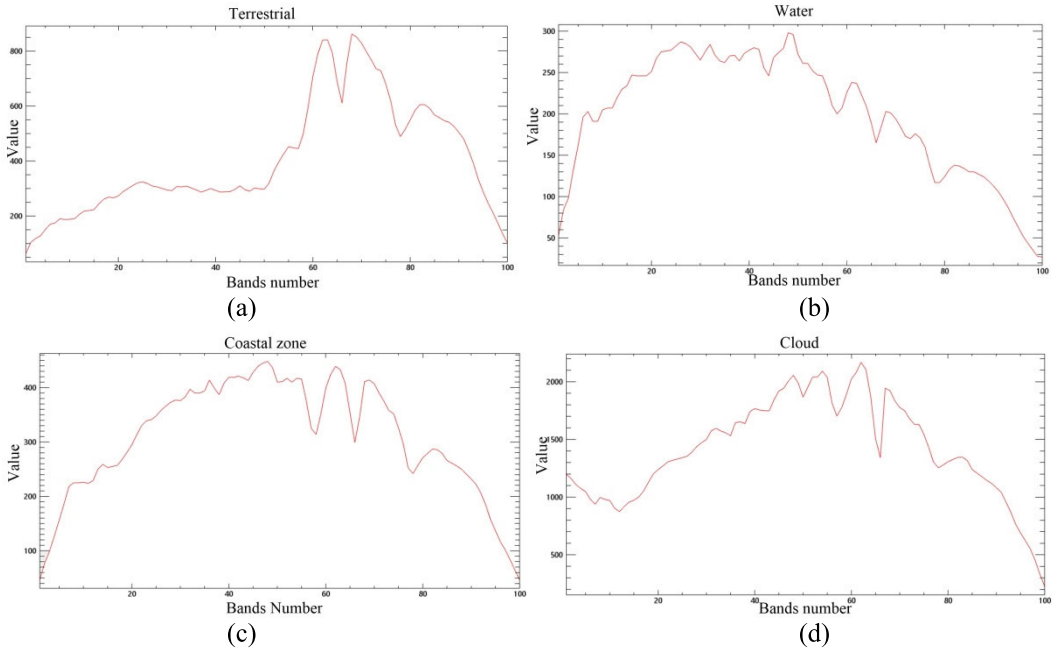


Fig. 10. Typical target spectral profiles. (a) Terrestrial, (b) waters, (c) coastal zone, and (d) cloud spectral response curve.

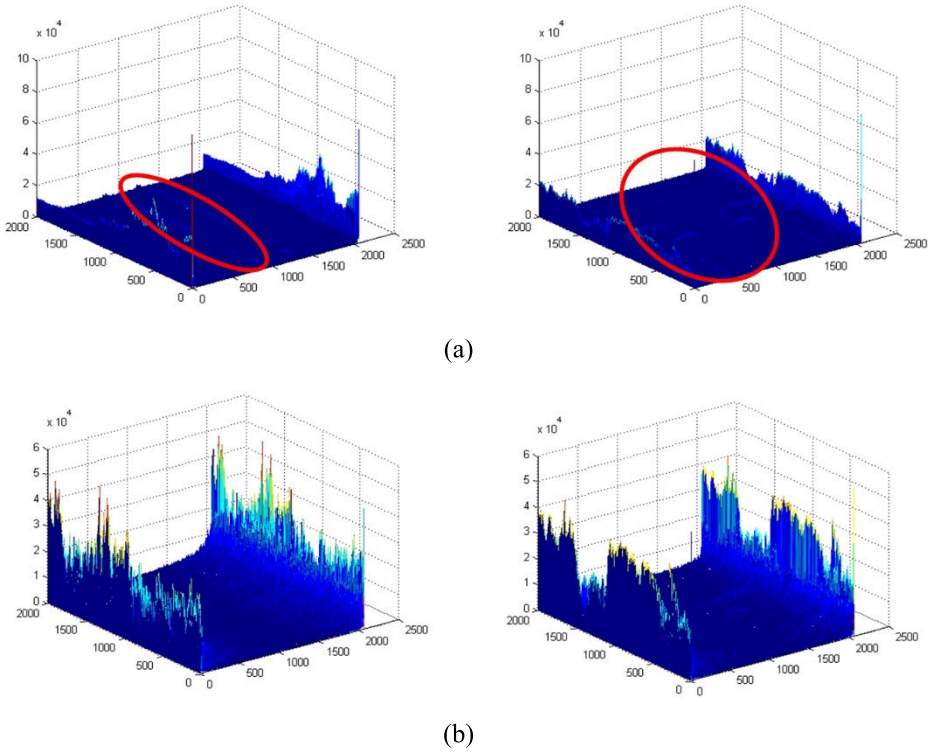


Fig. 11. Typical target spectral profile. Frequency domain image (a) with fringe noise and (b) without fringe noise.

Subsequently, PCA is adopted to decompose the images (see Fig. 12). As indicated by the decomposition results, the first three decomposed images cover most of the information, and the information in the later decomposed images is largely noise, the most typical of which is the vertical stripes. Thus, a PCA inversion is constructed to extract information from the first three bands, which in turn removes streaks and enables image enhancement.

B. Model Training Results

To verify the performance of the algorithm proposed in this article, we use ablation study to compare the algorithm performance of 3-D CNN, SENet, multilevel SENet (M-SENet), and the tandem structure of 3-D CNN and M-SENet proposed by us, as shown in Fig. 13. Fig. 13(a) presents the network *Loss* curves, with all networks displaying a decreasing trend. In the 3-D CNN network, the HSIs are input, with

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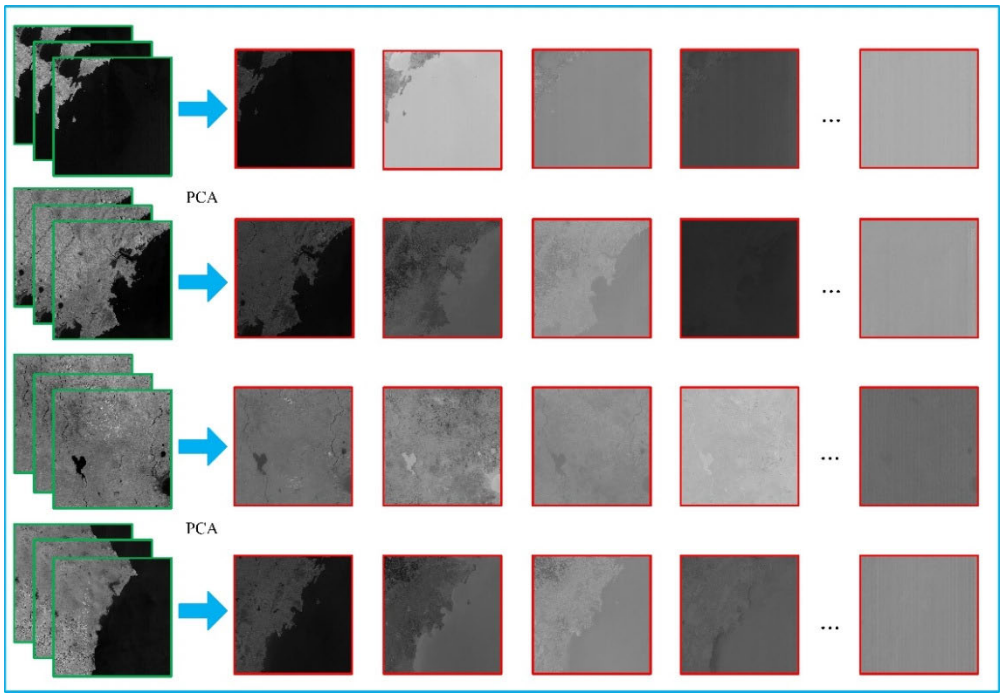


Fig. 12. PCA decomposition effects.

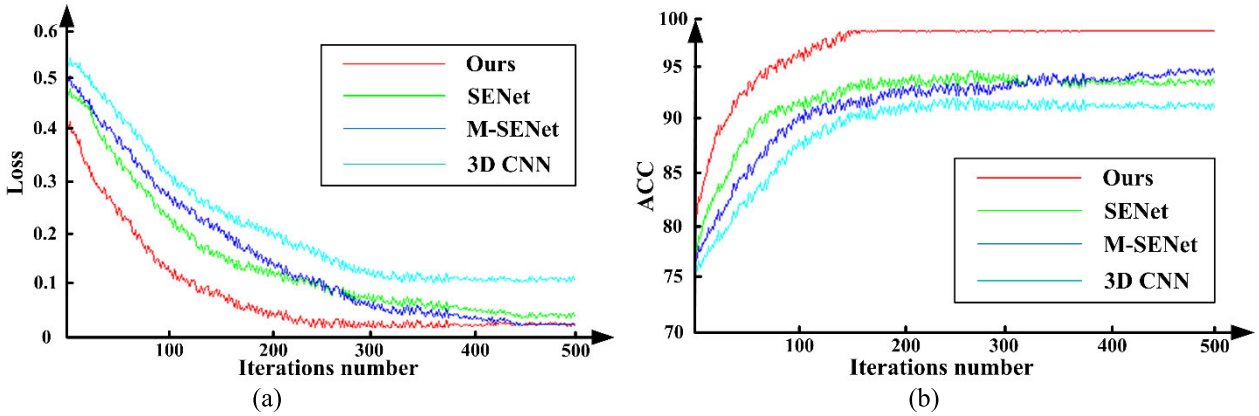


Fig. 13. Model training curves. (a) Loss curves. (b) Accuracy curves.

300 iterations. This network converges when the *Loss* reaches nearly 0.18. SENet introduces an attention mechanism to simulate visual perception, with a fast-decreasing *Loss* at early stages. Besides, it converges when the *Loss* reaches nearly 0.1 at 350 iterations. The M-SENet network introduces the idea of multiple levels on top of the SENet, leading to a slower decline in *Loss* at early stages compared with SENet, and a faster decline at later stages. Furthermore, it converges when the *Loss* reaches around 0.05 at 420 iterations. The proposed algorithm enhances the robustness of the network by concatenating the 3-D CNN with M-SENet, converging at a *Loss* of around 0.05 at 250 iterations.

As depicted in the accuracy curve in Fig. 13(b), the algorithm accuracy tends to increase with time, and the 3-D CNN, SENet, and M-SENet cannot be continuously improved when the accuracy reaches 0.95. Furthermore, the proposed network, which introduces attention mechanisms, multilevel features to build 3-D CNNs and a tandem structure of multilevel networks, achieves better results.

C. Algorithm Comparison

The HSI is illustrated in Fig. 14(a), with the shore on the left and the sea on the right of the image. Fig. 14(b) presents the corresponding real coastline, and the effect of the straightened coastal zone is presented in Fig. 14(c). The visible coastline is located in the middle of the image, showing a line form. The effect is examined by the level of the line. The higher the level, the better the effect will be.

The following metrics are introduced to analyze algorithm performance: accuracy rate (R_{acc}), indicating the ratio of positive samples correctly predicted to the total number of samples;

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(a) (b) (c)

Fig. 14. Coastal zone straightening effect. (a) Pseudo-color image by extracting three bands of hyperspectrum. (b) Real coastline. (c) Straightened coastal zone image.

precision (R_{prec}), indicating the ratio of the number of positive samples predicted by the classifier to the entire positive sample; sensitivity (R_{sen}), indicating the ratio of all positive samples correctly predicted by the classifier; specificity (R_{spe}), indicating the ratio of all negative samples correctly predicted by the classifier. Kappa (R_{Kappa}) coefficients are adopted to measure consistency testing to measure classification effectiveness

$$R_{\text{acc}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{22}$$

$$R_{\text{prec}} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{23}$$

$$R_{\text{sen}} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{24}$$

$$R_{\text{spe}} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{25}$$

$$R_{\text{Kappa}} = \frac{R_{\text{acc}} - p}{1 - p} \tag{26}$$

where TP denotes the number of pixels where the classifier predicts a positive sample, and actually also a positive sample. TN expresses the number of pixel points for which the classifier predicts negative samples and is actually negative as well. FP represents the number of pixels for which the classifier predicts a positive sample and is actually a negative sample. FN expresses the number of pixels for which the classifier predicts negative samples and is actually a positive sample.

Existing methods for coast zone segmentation have focused on conventional feature methods and depth feature methods. For conventional feature methods, Qiao et al. [37] proposed NDWI using a specific threshold (NDWI-T) based on the ratio between the infrared and visible bands for the extraction of water bodies. This method, analyzed in principle, exhibits better performance for boundaries where the sea and shore are more clearly distinguished, especially for the extraction of the demarcation between deep water and shore; it is less effective for the segmentation of the coastal buffer zone. In literature [26], the extraction of coastline was performed through the textural features (*Texture*) of the water body showing ripples. This method needs a high image resolution while requiring the water to reflect only the surface information, which is less effective when the seawater is clear but the seafloor is noisy. In literature [55], the energy distribution characteristics of hyperspectral features and water bodies were analyzed. As several spectral bands are severely attenuated in water bodies, cross-entropy can be built from the energy perspective and active contour (AC) can be built from the boundary perspective for the segmentation of coastlines. The above-mentioned method combines the energy and boundary characteristics of hyperspectral features and exerts a better segmentation effect than the previous two methods.

For depth feature methods, the literature [42] built deep learning-based mask R-CNN to achieve coastline extraction from the perspective of classification. However, only limited waveband information has been considered for building the model for coastline extraction, and the information on hyperspectral has not been fully utilized. DAU-Net [44] built relations between images and semantics to achieve coastline extraction with some effectiveness. CAN [45] incorporates attention into the network to simulate visual areas of interest for coastline extraction. The above algorithms build models in a single dimension for the segmentation of coast zones, whereas they fail to fully exploit the characteristics of high spectral resolution and high spatial resolution of remote sensing images. These algorithms still have room for improvement. Besides, the algorithm proposed in this study, in accordance with the principle of visual perception, fully exploits the characteristics of HSIs, while a model is built from spatial and interspectral dimensions. An attention-based 3-D CNN is proposed, the SENet model is built based on multilayer features from the multiscale perspective, and a tandem structure is adopted to fuse the two to achieve the extraction of the coast zone with better results.

Four common scenarios were created and put to the test. The long coastal zone extraction findings are displayed in Table I, the inland coastal zone extraction findings are displayed in Table II, far-sighted long coastal zone extraction findings are displayed in Table III, and near-view long coastal zone and island coastal zone extraction findings are displayed in Table IV. It can be seen that the long coastal zone has the best extraction effect, while the effect of inland coastal zone, sea long coastal zone, and island coastal zone decreases. Moreover, extractions of close-up long coastal zone and island coastal zone are the most challenging and inefficient.

Fig. 15 presents the results of the conventional feature method AC, the deep learning method CAN, and the proposed algorithm. Fig. 15(a) illustrates the pseudo-color image by extracting three bands of hyperspectrum. The AC algorithm misidentifies the coastline due to the presence of suspended material on the seafloor, which leads to biased segmentation as Fig. 15(b), and CAN uses a uniform scale to

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TABLE I
LONG COASTAL ZONE EXTRACTION

Algorithm	R_{acc}	R_{prec}	R_{sen}	R_{spe}	R_{kappa}
NDWI-T	0.81	0.71	0.79	0.76	0.72
Texture	0.84	0.78	0.83	0.81	0.79
AC	0.87	0.81	0.85	0.84	0.83
Mask R-CNN	0.85	0.87	0.88	0.86	0.82
DAU-Net	0.93	0.86	0.91	0.92	0.86
CAN	0.95	0.91	0.93	0.92	0.89
Ours	0.98	0.95	0.96	0.93	0.94

TABLE II
INLAND COASTAL ZONE EXTRACTION

Algorithm	R_{acc}	R_{prec}	R_{sen}	R_{spe}	R_{kappa}
NDWI-T	0.75	0.73	0.74	0.71	0.72
Texture	0.79	0.75	0.78	0.73	0.74
AC	0.84	0.76	0.76	0.78	0.76
Mask R-CNN	0.86	0.81	0.82	0.81	0.78
DAU-Net	0.85	0.84	0.85	0.85	0.82
CAN	0.89	0.87	0.88	0.89	0.86
Ours	0.94	0.93	0.91	0.92	0.91

TABLE III
VISIONARY LONG COASTAL ZONE AND ISLAND COASTAL ZONE EXTRACTIONS

Algorithm	R_{acc}	R_{prec}	R_{sen}	R_{spe}	R_{kappa}
NDWI-T	0.75	0.73	0.72	0.71	0.68
Texture	0.79	0.75	0.74	0.72	0.71
AC	0.81	0.77	0.78	0.75	0.75
Mask R-CNN	0.83	0.81	0.79	0.79	0.76
DAU-Net	0.86	0.84	0.83	0.82	0.81
CAN	0.88	0.86	0.84	0.85	0.86
Ours	0.91	0.89	0.88	0.91	0.89

TABLE IV
CLOSE-UP LONG COASTAL ZONE AND ISLAND COASTAL ZONE EXTRACTIONS

Algorithm	R_{acc}	R_{prec}	R_{sen}	R_{spe}	R_{kappa}
NDWI-T	0.75	0.70	0.68	0.71	0.72
Texture	0.78	0.71	0.71	0.73	0.74
AC	0.83	0.74	0.72	0.75	0.77
Mask R-CNN	0.81	0.77	0.76	0.79	0.80
DAU-Net	0.86	0.81	0.78	0.82	0.82
CAN	0.88	0.85	0.82	0.86	0.84
Ours	0.89	0.87	0.85	0.89	0.87

analyze the areas with sharp turns, which are incorrectly segmented and focused to the inner boundary as Fig. 15(c). However, the algorithm proposed in this study builds a model from spatial, interspectral, and multiscale perspectives to

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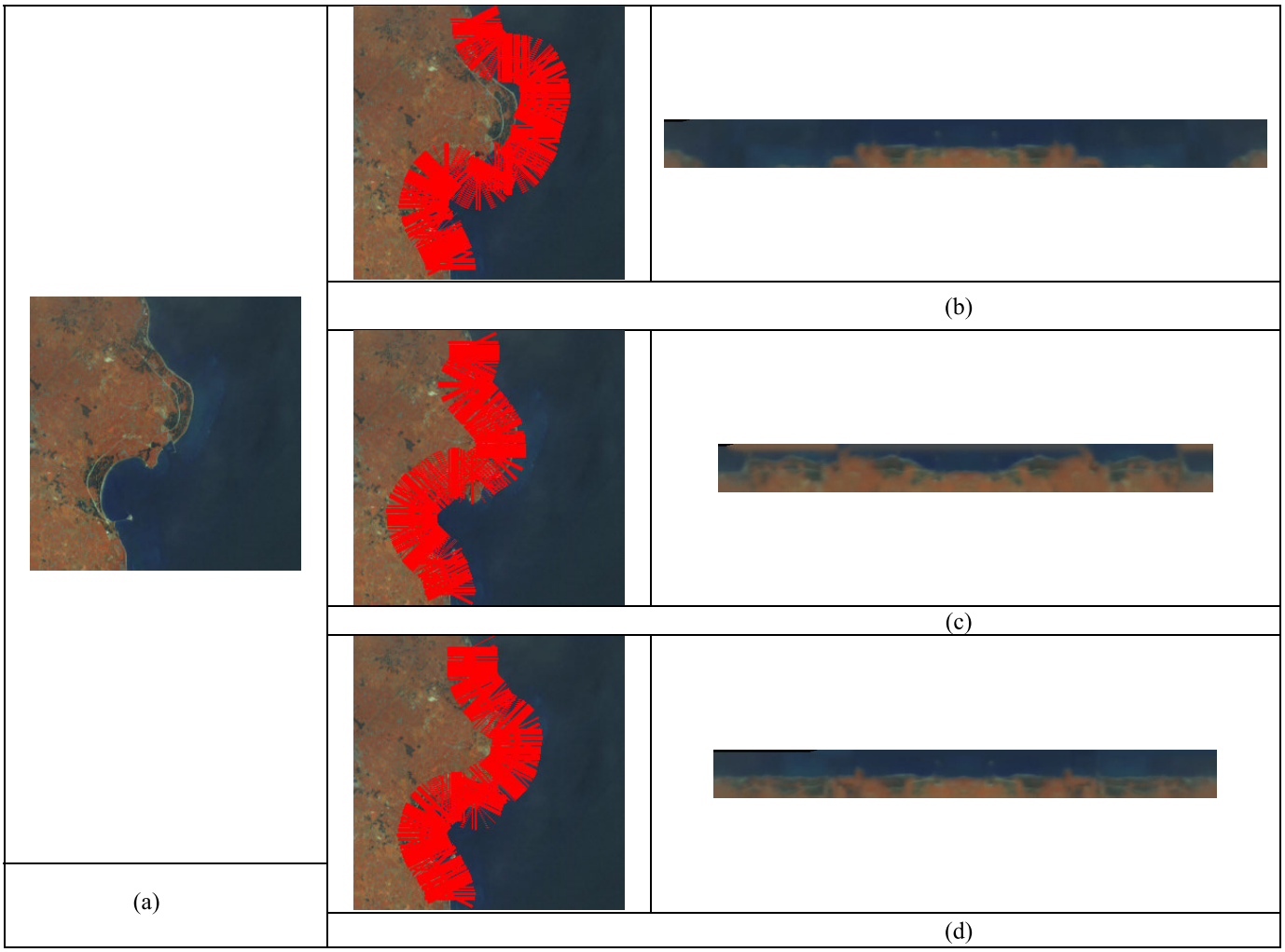


Fig. 15. Result images by different algorithms. (a) Pseudo-color image by extracting three bands of hyperspectrum. (b) AC algorithm result. (c) CAN algorithm result. (d) Our algorithm result.

straighten the coastal strip, and the straightening effect shows that the dividing line between sea and shore is in the middle of the image, which verifies the effectiveness of the algorithm as Fig. 15(d).

D. Result Presentation

To verify the effectiveness of the algorithm proposed in this study, we selected representative coast zones to be analyzed. Fig. 16(a) shows a clear distribution of land and sea, with land on the left of the image and sea on the right, and the coast zone is smoother and better extracted. As can be seen in the straightened images, there is a section in the early part of the image showing the water body both above and below, which is due to the substantial turn of the coastline. The visible image in Fig. 16(b) consists of inland lakes and nearshore shallows, with the inland lakes showing a closed form and the nearshore shallows showing a gradual transition. The straightening effect shows that the inland lakes are better, while the shallows are not clearly delineated, resulting in blurred images. The visible image from Fig. 16(c) contains a large area of water containing a long coastline and an island. The large degree of curvature of the long coast in this scene, combined with the distance, makes the image blurred and extraction more difficult. The island is in the water body and forms an enclosed area with a better straightening effect. Fig. 16(d) is the corresponding close-up image of Fig. 16(c), containing three separate coast zones. The long coastal zone image has a more pronounced straightening effect and is clearer in detail. The general pattern of the islands is consistent with Fig. 16(c) and shows more detail. In summary, the algorithm proposed in this article has strong robustness and good effect for different coastal zones.

The problem has been identified in the image straightening process, where the coastline was selected as the center and the perpendiculars of two adjacent points were chosen to be straightened at a specified radius. Nevertheless, due to differences in the angle of the perpendiculars, null values were reported in the straightening process as shown in Fig. 17(a). We used mean filtering as shown in Fig. 17(b) to fix this issue, but it reduced the image resolution and the images were blurred. Further research based on this issue will follow.

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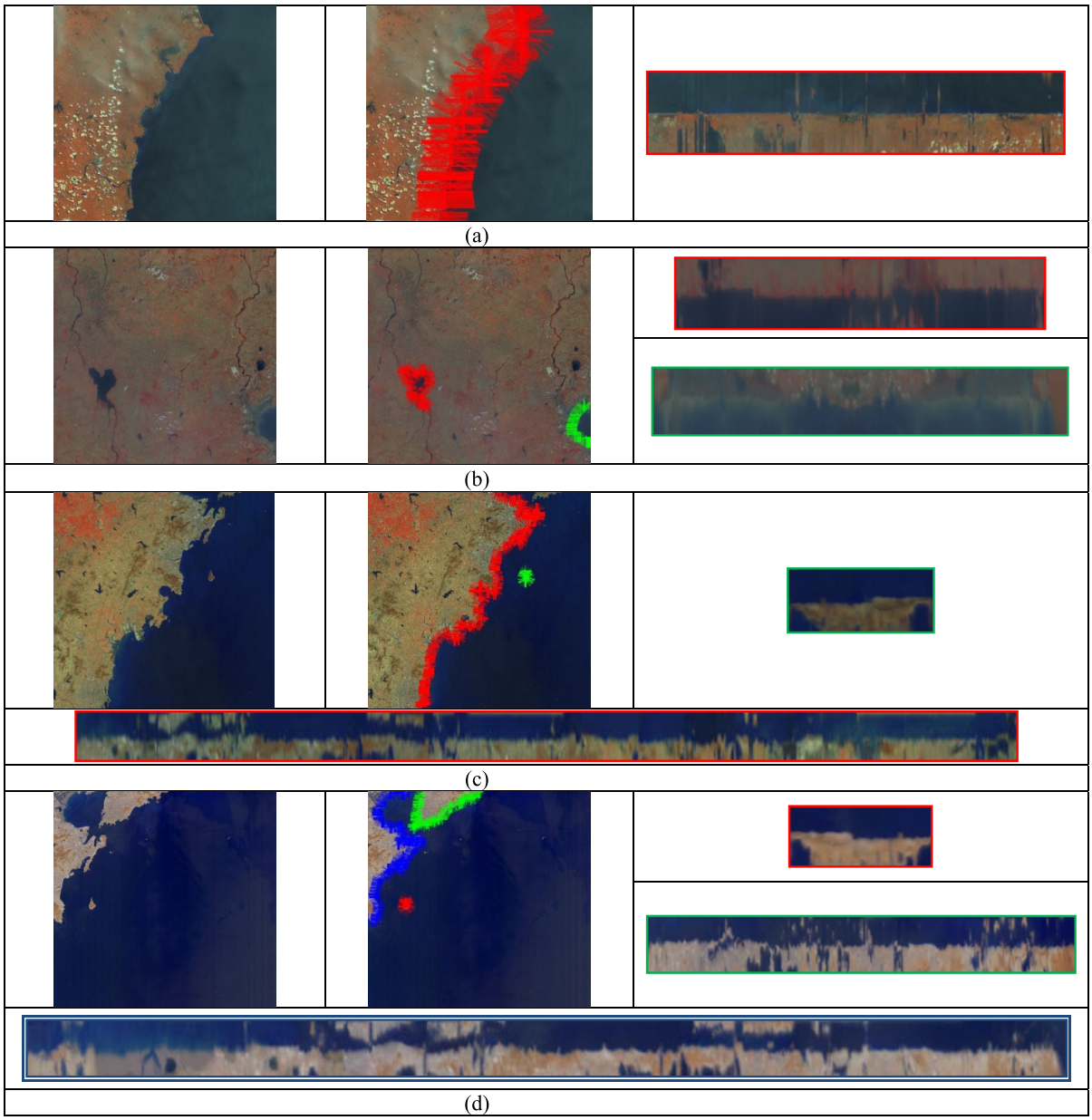


Fig. 16. Coastline straightening effects. (a) Long coastal strip straightening effects. (b) Inland coastal zone straightening effects. (c) Long coastal strip and island coastal strip straightening effects in the foreground. (d) Close-up long coastal strip and the straightened effects of the island coastal strip.

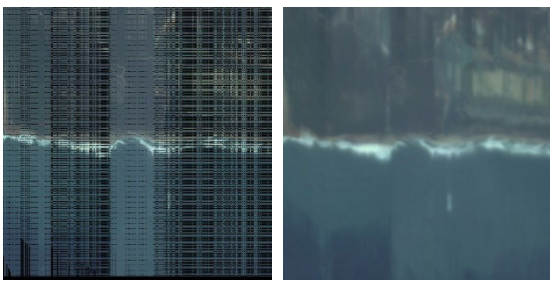


Fig. 17. Problems with coastline straightening. (a) Direct mapping. (b) Uniform filtering.

IV. Conclusion

The coastal zone refers to the boundary between water bodies and land, and it exhibits both water and land properties. A model is built from HSIs for image preprocessing and feature extraction, and a coastal zone extraction algorithm is proposed for HSIs based on multilayer depth features. Specific innovations are presented below: 1) the PCA image enhancement algorithm is proposed through the analysis of HSI noise; 2) the visual principles are simulated to develop attention mechanisms that place a focus on the coastal zone area; and 3) a multifeature bearer model is proposed, fusing 3-D CNN and SENet, and introducing a multiscale mechanism to accommodate coastal zone extraction at different scales. A coastal strip straightening is employed in this study to visualize the effect of the algorithm.

Several limitations remain in this study as follows: 1) HSIs contain considerable bands and a large amount of data. In general, when inputting images to conduct research based on deep learning networks, more time and computational resources

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will be inevitably consumed, and the need for rapid detection cannot be satisfied. 2) Some variability exists in the energy displayed in HSIs in different spectral bands, such that a weak and under-utilized image signal is generated in the starting bands. 3) The limited spectral data that have been acquired for the coastal zone makes it difficult to investigate typical target detection and mining. Furthermore, given the above limitations, subsequent research will be conducted.

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